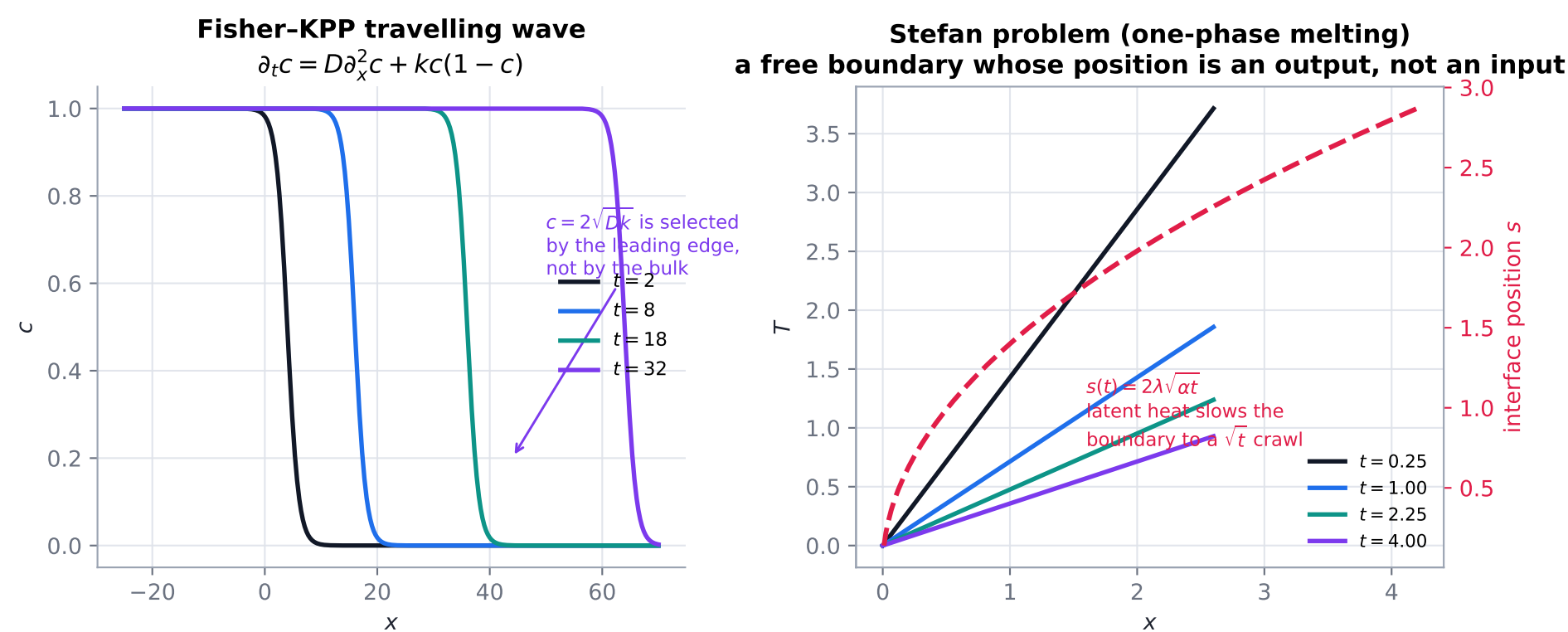
Fisher–KPP travelling wave
$\partial_t c = D\partial_x^2 c + kc(1-c)$
c
x
$c = 2\sqrt{Dk}$ is selected
by the leading edge,
not by the bulk
$t = 2$
$t = 8$
$t = 18$
$t = 32$
Stefan problem (one-phase melting)
a free boundary whose position is an output, not an input
T
x
interface position s
$s(t) = 2\lambda\sqrt{\alpha t}$
latent heat slows the
boundary to a $\sqrt{t}$ crawl
$t = 0.25$
$t = 1.00$
$t = 2.25$
$t = 4.00$